

Laboratory Report: Advanced Scope Measurements

Experimentation and Analysis of Signal Coupling, Pulse Characteristics, and Diode I-U Behavior

1. Objectives

- Analyze the impact of AC vs. DC coupling on measurement accuracy and waveform representation.
- Determine pulse timing parameters including rise time, fall time, and duty cycle.
- Utilize X-Y mode to characterize the non-linear I-U curve of a 1N4007 Silicon Diode.

2. Experiment 1: Impact of DC/AC Coupling

The frequency response of a signal was measured under both DC and AC coupling modes to observe the high-pass filter effect inherent in AC coupling.

Amplitude Response Data

Frequency (Hz)	U1 (DC) [V]	U2 (AC) [V]	Gain (dB)
1	2.00	2.00	0.00
10	2.00	2.00	0.00
30	1.00	2.00	+6.02
50	1.00	2.00	+6.02

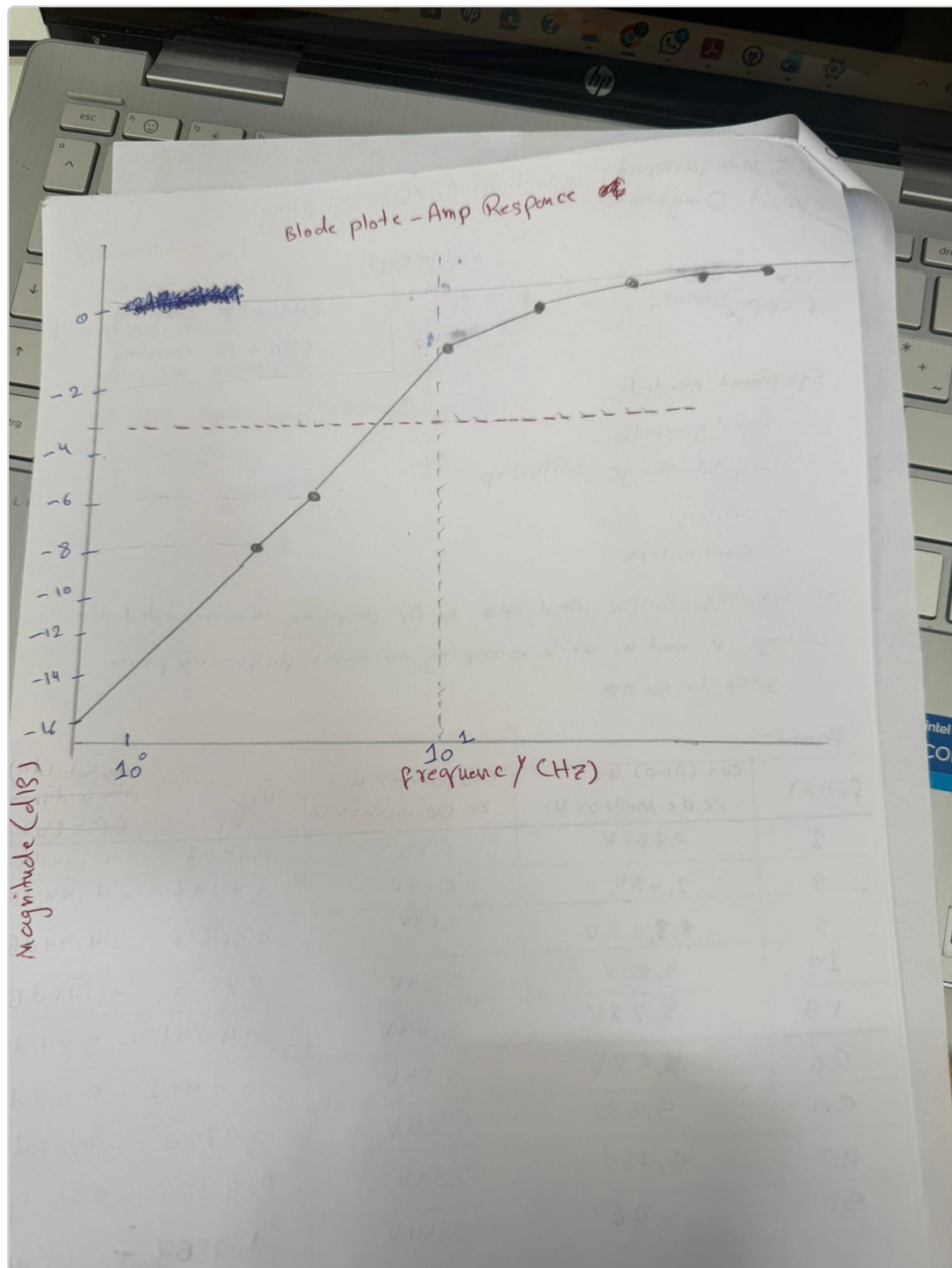


Figure 1: Handwritten Bode Plot showing the amplitude response over frequency.

3. Experiment 2: Pulse Characteristics

A square wave pulse was analyzed to determine the switching speeds of the generator and the signal's duty cycle.

Measured Pulse Parameters:

- **Amplitude:** 4.64 V
- **Mean Voltage:** 3.61 V
- **Rise Time (t_R):** 211.8 μ s
- **Fall Time (t_F):** 79.50 μ s
- **Calculated Duty Cycle:** 77.8%

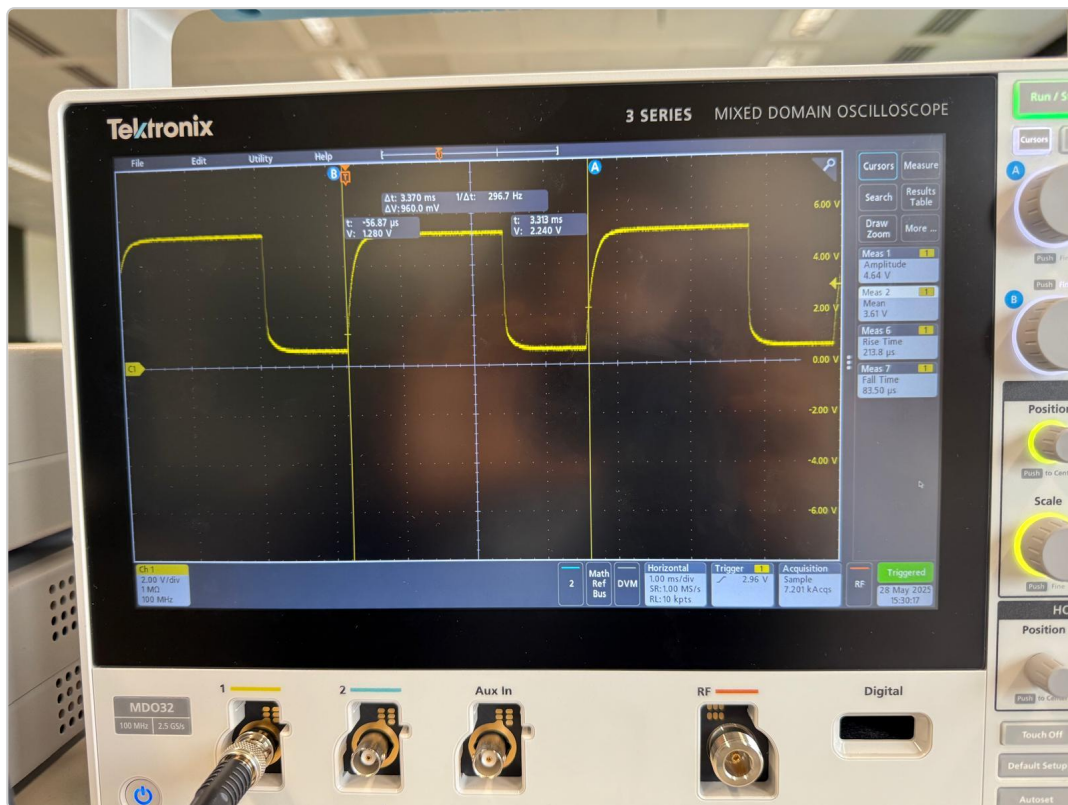


Figure 2: Oscilloscope capture of the pulse signal measurement.

4. Experiment 3: X-Y Mode & Diode Characterization

Using a differential probe on Channel 2 (Current) and Channel 1 (Voltage), the characteristic curve of a Silicon Diode (1N4007) was plotted in X-Y mode.

Theoretical Calculation

The current I was calculated using the diode equation: $I = I_s (e^{U/(n \cdot U_T)} - 1)$.

- At 600mV: $I \approx 1.025 \text{ mA}$
- At 800mV: $I \approx 480.01 \text{ mA}$

⑧ Diode is electronic component that allows current to flow in only one direction, its purpose is to convert AC to DC current or DC and protect circuits from reverse voltage as well as control the direction of current flow in electric devices.

silicium diode IV diagram range 0-800mV by $I = I_s (e^{\frac{V}{n \cdot U_T}} - 1)$

$$I_s = 10 \text{ nA} \Rightarrow 10 \cdot 10^{-9} \text{ A}$$

$$n = 2$$

$$U_T = 26 \text{ mV} = 26 \cdot 10^{-3} = 0,026 \text{ V}$$

$$V = 600 \text{ mV} \Rightarrow 600 \cdot 10^{-3} = 600 \cdot 0,001 = 0,6$$

↳ we want to check this voltage causes how much current.

$$I = 10 \cdot 10^{-9} \left(e^{\frac{0,6}{2 \cdot 0,026}} - 1 \right) = 10 \cdot 10^{-9} \left(e^{\frac{0,6}{0,052}} - 1 \right) =$$

$$10 \cdot 10^{-9} \left(e^{11,538} \right) = 10 \cdot 10^{-9} \left(2,71828^{11,538} \right) =$$

$$10 \cdot 10^{-9} \cdot 102538359 = 0,00102538359 \text{ A} \Rightarrow \boxed{1,025 \text{ mA}}$$

→ at 600mV the current through the diode is approximately 1,025 mA.

we also calculate it for 800mV $= 800 \cdot 10^{-3} = 0,8 \text{ V}$

$$I = I_s \left(e^{\frac{V}{n \cdot U_T}} - 1 \right) = 10^{-8} \left(e^{\frac{0,8}{2 \cdot 0,026}} - 1 \right) = 10^{-8} \left(e^{15,3846} - 1 \right)$$

$$= 10^{-8} (480184,27 - 1) = 10^{-8} (480183,27) =$$

$$\Rightarrow \boxed{480,01 \text{ mA}}$$

we can compare it now that at 600mV the current was 1,025 mA but at 800mV the current suddenly increased to 480,01 mA.

Figure 3: Theoretical diode current calculations at specific voltage points.

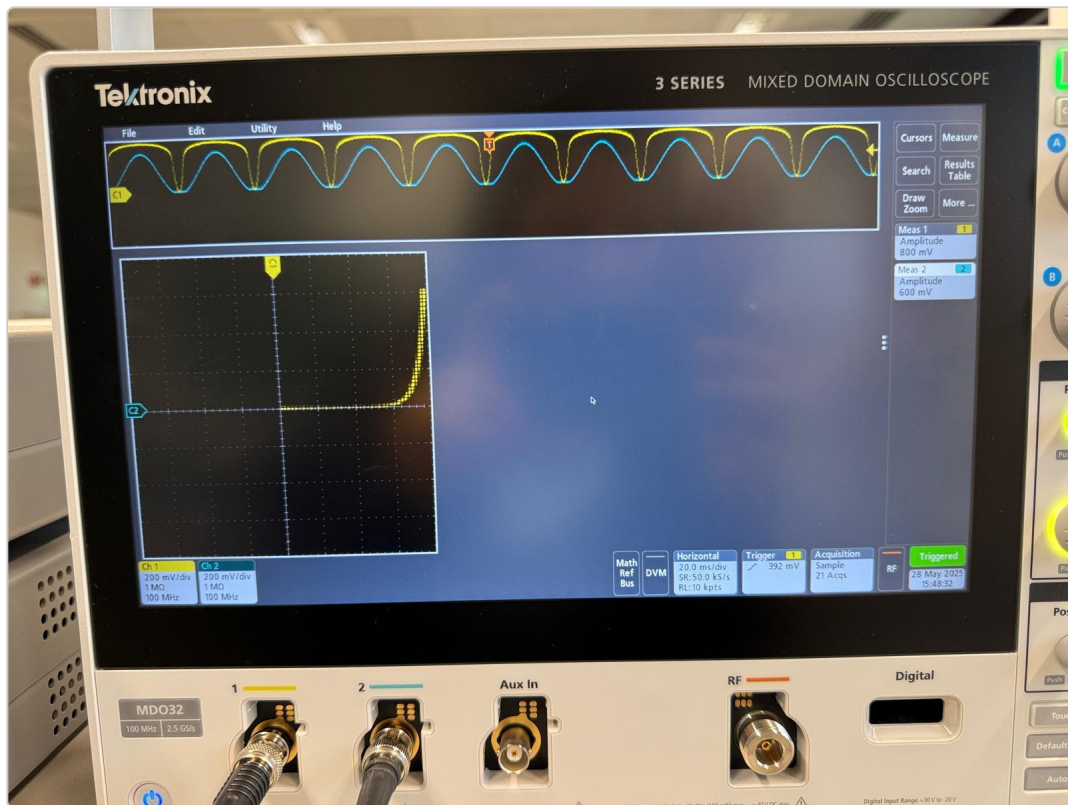


Figure 4: Real-time I-U characteristic of the 1N4007 diode displayed in X-Y mode.

5. Conclusion

The experiments successfully demonstrated the practical applications of advanced oscilloscope functions. AC coupling was confirmed to block DC components but also introduce attenuation at low frequencies. Pulse measurements highlighted the asymmetry in rise and fall times, while X-Y mode provided a direct visual correlation between diode voltage and exponential current growth.